THE CURVES STACK IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition artin-definition-limit-preserving (canonical tag 07XL). cited at statement line 4877. The property being established.
- Equation equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4881. The embedding of the curves stack.
- Lemma lemma-spaces-limits (canonical tag 0D1I). cited at proof line 4884. The spaces stack is limit preserving.
- Lemma spaces-limits-lemma-eventually-relative-dimension (canonical tag 0D4K). cited at proof line 4896. Relative dimension <= 1 eventually.

Source reference. Original definition 07XL: sources/stacks-chow/tags.txt, source lines 10081–10081. Cited at statement line 4877: the property being established.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at proof line 4881: the embedding.

Source reference. Original lemma 0D1I: sources/stacks-chow/tags.txt, source lines 16348–16348. Cited at proof line 4884: the spaces stack fact.

Source reference. Original lemma 0D4K: sources/stacks-chow/tags.txt, source lines 16455–16455. Cited at proof line 4896: the finite-stage statement.

Lemma 0.2. *The stack $\mathcal{C}urves \to Sch_{fppf}$ is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Using the embedding (0D50), the description of the image, and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1I) this reduces to the following statement: Let $T = \lim T_i$ be the limits of a directed inverse system of affine schemes. Let $i \in I$ and let $X_i \to T_i$ be an object of $\mathcal{S}paces'_{fp,flat,proper}$ over T_i. Assume that $T \times_{T_i} X_i \to T$ has relative dimension ≤ 1. Then for some $i' \geq i$ the morphism $T_{i'} \times_{T_i} X_i \to T_i$ has relative dimension ≤ 1. This follows from Limits of Spaces, Lemma 0D4K. □

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